

Esther Thompson

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EDUCATION

Bachelor of Science, Computer Science

September 2022 – December 2026

University of Calgary, Alberta

- **Awards/Accomplishment:** Science Internship Program and Calgary Flames Foundation Outstanding Youth Scholarship
- **Relevant courses:** Software Engineering, Data Structures and Algorithms, Database Management Systems, Human Computer Interactions, Computer Networks, and Explorations in Machine Learning

EXPERIENCE

Girls Coding Instructor

May 2023 – Present

Minds in Motion - University of Calgary

- Developed detailed teaching content and documentation through research of relevant topics to ensure an effective learning experience and assist instructors with onboarding and troubleshooting.
- Taught students core programming principles through hands-on exercises that utilized both block-based and text-based tools, such as Scratch, Python, and JavaScript. As a result, 90% of students reported increased confidence in their coding abilities.
- Simplified complex topics such as data structures and algorithms for diverse learning styles, resulting in significantly higher student participation and enthusiasm.

VP Communications

May 2024 – Present

Tech Start - University of Calgary

- Launched targeted social media campaigns and designed promotional materials in Figma, resulting in increased member engagement and a significant boost in recruitment.

PROJECTS

Palette Genie | [GitHub](#) | Typescript, Node.js, Python, TensorFlow, MySQL, Hapi.js

June 2025 – Present

- Developing an AI-powered web application that calculates accurate mixing ratios from users' personal paint inventories to replicate reference colours from photos.
- Applying backend services to process images and perform machine learning inference for accurate color matching.
- Designing secure data storage and authentication systems to support personalized user experiences.

Whisk-Y Baking | [GitHub](#) | JavaScript, Node.js, Supabase, PostgreSQL, CSS

March 2025 – April 2025

- Developed a full-stack web app with a card-based layout and smart filtering to help users browse, save, and find recipes based on ingredients, dietary needs, and personal preferences.
- Implemented user authentication, profile management, and recipe upload features, allowing users to personalize their experience and contribute content.
- Created the relational database schema, wrote complex SQL queries, and handled database design, query optimization, and data integrity.

TECHNICAL SKILLS

Languages: Python, JavaScript, Typescript, Java, C, C++, C#, Swift, SQL, PostgreSQL, HTML, CSS

Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, Hapi.js, Tailwind CSS

Developer Tools: GitHub, VS Code, Xcode, Jira, Postman, Docker, MongoDB, Supabase, AWS, Microsoft Azure, Cursor, Claude AI, Microsoft Copilot, JUnit, Cucumber, Power BI